

ran-
dom
per-
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space

$$S = \{0, 1, \dots, 9\}$$

ran-
dom
event
 $A \subset$

$$A = \{0, 2, 4, 6, 8\} \subset S = \{0, 1, \dots, 9\}$$

$s \in A$
 $P(A)$

$$0 \leq P(A) \leq 1$$

$$P(\emptyset) = 0$$

$$A = \{0, 1, 2, 3\} \subset S$$

$$P(A) = 4/10 = 0.4$$

(S, A, P)

prob-
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tion

$$A \cup B$$

$$A \cap B$$

$$A \cap B = \emptyset$$

$$P(A \cap B) = 0$$

Union
 $A \cup B$
 $A \cap B$
 $A \cap B$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

(1)

$$P(A \cup B) = P(A) + P(B)$$

(2)

com-
ple-
ment

$$A^C = S \setminus A$$

$$A^C = S \setminus A$$